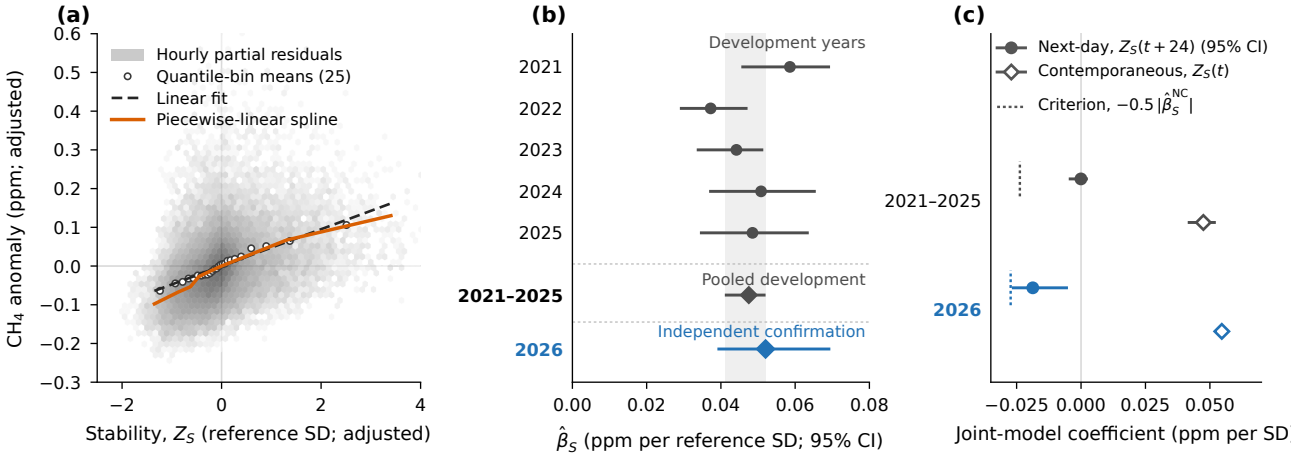

(a)
Hourly partial residuals
Quantile-bin means (25)
Linear fit
Piecewise-linear spline
CH4 anomaly (ppm; adjusted)
Stability, ZS (reference SD; adjusted)
(b)
Development years
2021
2022
2023
2024
2025
Pooled development
2021–2025
Independent confirmation
2026
β̂S (ppm per reference SD; 95% CI)
(c)
Next-day, ZS(t + 24) (95% CI)
Contemporaneous, ZS(t)
Criterion, −0.5 |β̂S^NC|
2021–2025
2026
Joint-model coefficient (ppm per SD)